

Competing Risks and Multi-State Models

Concepts, methods and software

Course given at *Karolinska Institute*

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Outline

Day 1; Competing risks, basic methods Examples. The main quantities. Data structure. Estimation and regression: cause-specific hazard

Day 2; Regression Estimation and regression: subdistribution hazard. Type of study questions: inferential, predictive, causal. Which model to use?

Day 3; Multi-state models Objectives. Terminology. Data. Non-parametric estimation.

Day 4; Prediction; Interpretation Prediction in multi-state models. Interpretation; when to choose which quantity?

Part I

Main Concepts and Quantities



Contents:

1. Examples
2. Intervention versus biological competing event
3. Main time-to-event quantities
4. Multi-state and subdistribution approach
5. Marginal or cause-specific distribution
6. Data structure

Outline

Examples Competing Risks

- All event types of interest

- One event type of interest, intervention as competing risk

- One event type of interest, biological event as competing risk

On Rates and Risks

- Classical survival analysis

- Competing risks analysis

- Multi-state approach

- Subdistribution approach

- Type of estimands in the presence of competing risks

Data structure

- Time scales

- Right censored data

- Late entry/left truncation

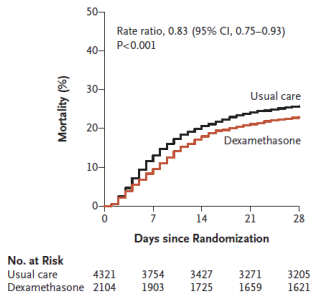
- Setup data analysis

Event-type outcomes that are “competing” (mutually exclusive)

- Each event type is of interest
 - A: Effect of dexamethasone on recovery and death after SARS-CoV-2 infection (binary outcome, complimentary levels)
 - B: Spectrum in causes of death after HIV infection (role of cART)
 - C: AIDS and SI switch after HIV infection
 - D: Death and relapse after bone marrow transplant (Practical)

A: Recovery (dexamethasone in Covid-19) trial

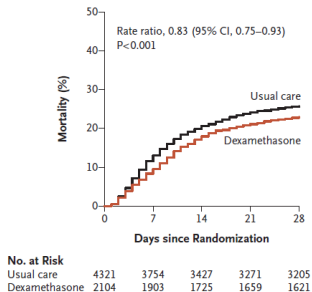
A All Participants (N=6425)



- Primary outcome: all-cause mortality within 28 days
- Secondary outcome: hospital discharge $\leq$ 28 days
- Binary outcome: death and recovery complimentary outcomes $\rightarrow$ logistic regression

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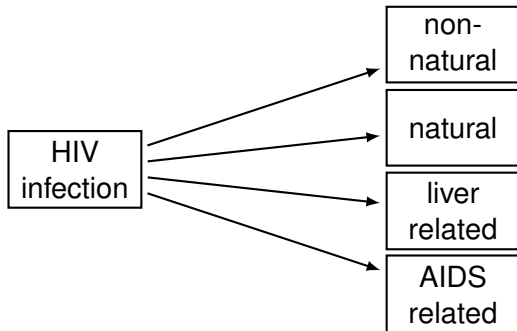


- Primary outcome: all-cause mortality within 28 days
- Secondary outcome: hospital discharge $\leq$ 28 days
- Binary outcome: death and recovery complimentary outcomes $\rightarrow$ logistic regression
- But they used time-to-event methods
 - 10% event-free in hospital after 28 days
 - event-free not a separate outcome
 - time to event may be of interest as well

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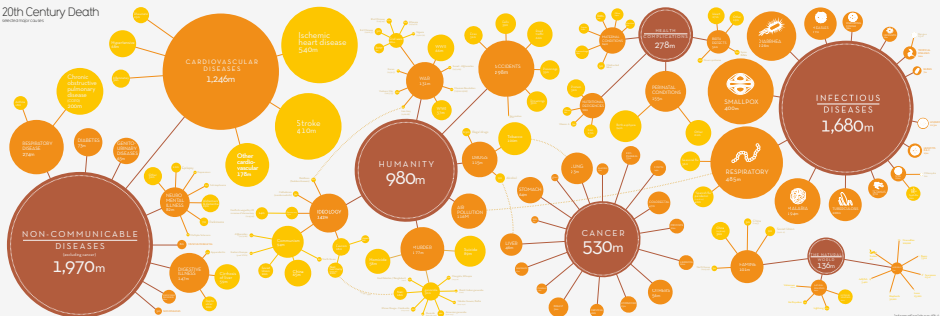
B: Spectrum in causes of death after HIV infection



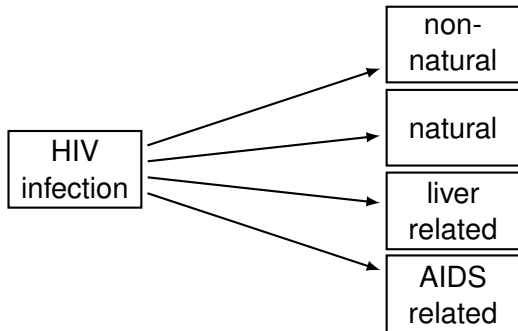
- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)

20th Century Death

selected major causes

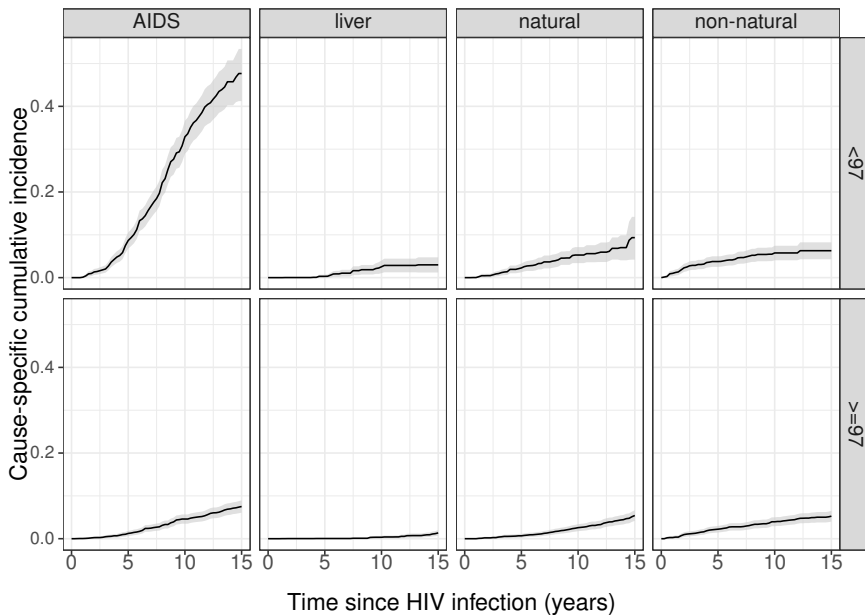


B: Spectrum in causes of death after HIV infection



- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)
- Side effects of cART? *Etiology* ($\approx$ cause-specific hazard)
- In the end we all die. *Prediction* ($\approx$ subdistribution hazard)

Cause-specific mortality by calendar period



Event-type outcomes that are “competing” (mutually exclusive)

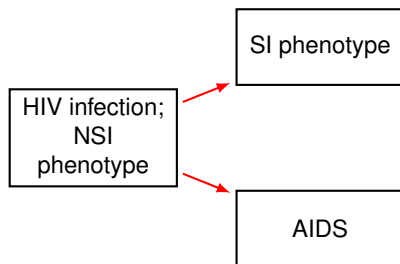
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C: AIDS and SI switch after HIV infection

HIV infection;
NSI
phenotype

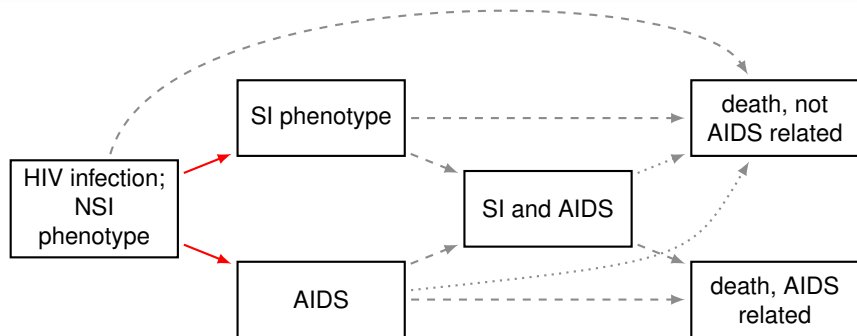
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C: AIDS and SI switch after HIV infection



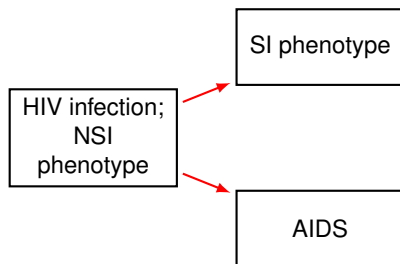
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- Competing risks: time to either SI switch or AIDS as first event
- Multi state model also describes what happens after first event

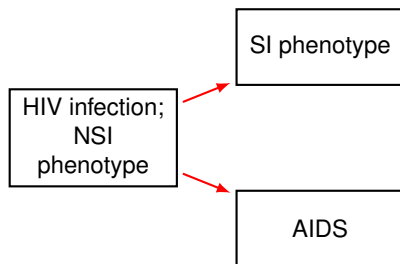
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- NSI virus needs CCR5 co-receptor for cell entry
- Gene coding for CCR5 may have deletion CCR5- Δ 32 on one chromosome
- Individuals with CCR5- Δ 32 have slower AIDS progression

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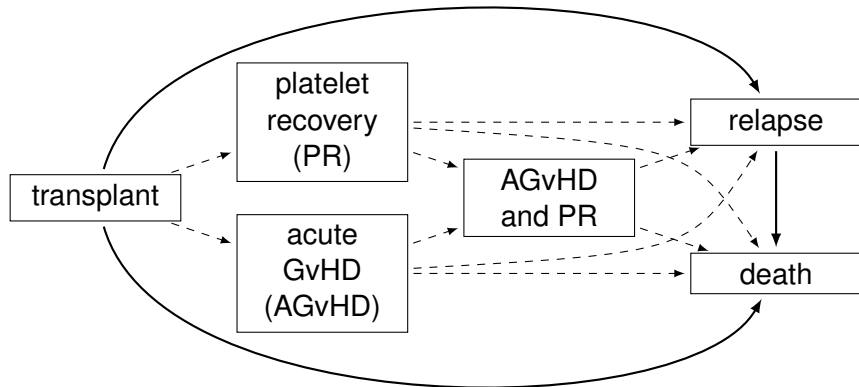
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- Question: SI switch as first event more likely for CCR5- Δ 32 individuals?

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D: Events after bone marrow transplantation



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Classical survival analysis

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Multi-state approach

Subdistribution approach

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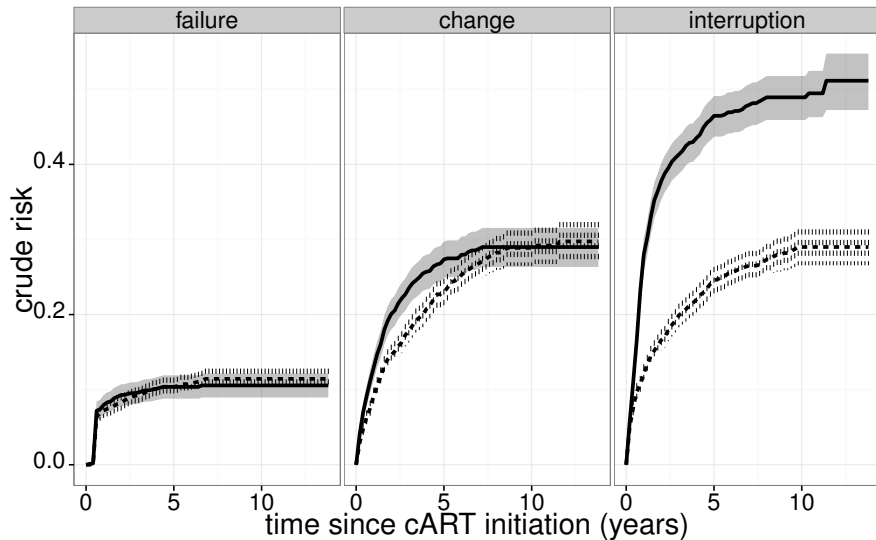
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 - E: Intervention (transplantation or treatment) competing risk for natural disease course
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 - G: Discharge competing risk for hospital-acquired infection

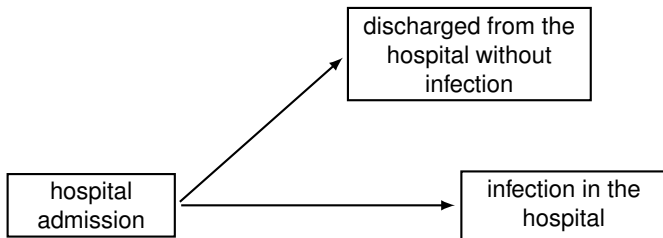
F: cART treatment failure; early (solid) versus late (dashed) starters



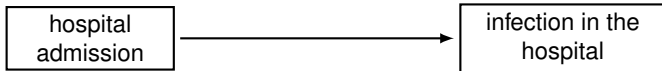
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G: Staphylococcus infection during hospital stay

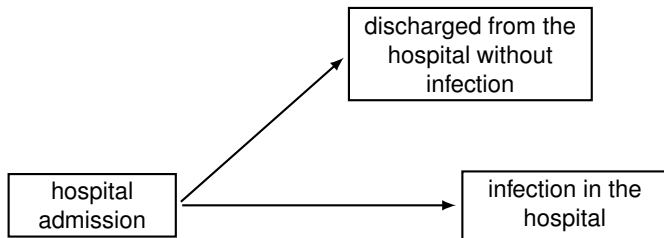


G: Staphylococcus infection during hospital stay



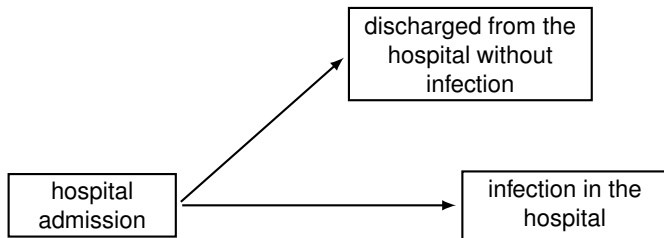
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 - marginal distribution/net risk

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 - cause-specific cumulative incidence /subdistribution/crude risk

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Estimation

week	0-1	1-2	2-3	3-4	4-5	5-6	6-7	> 7
infection	1	2	6	11	9	11	2	18
cumulative	1	3	9	20	29	40	42	60
discharge	5	9	6	6	9	12	4	35
cumulative	5	14	20	26	35	47	51	86

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- **Subdistribution.** Estimated as frequency of events:

$$\hat{P}(\text{infection} \leq 6 \text{ weeks}) = 40/146$$

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Individuals with competing event remain in denominator,
competing event ignored in estimation

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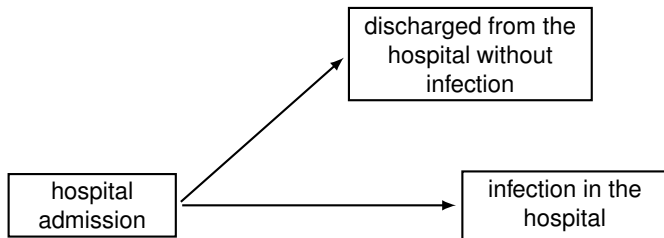
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- Marginal distribution.** Discharged individuals treated and interpreted as censored; Kaplan-Meier nonparametric estimator, assumes independent censoring

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 - marginal distribution/net risk
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- Which of two hospitals has higher risk may depend on type of risk

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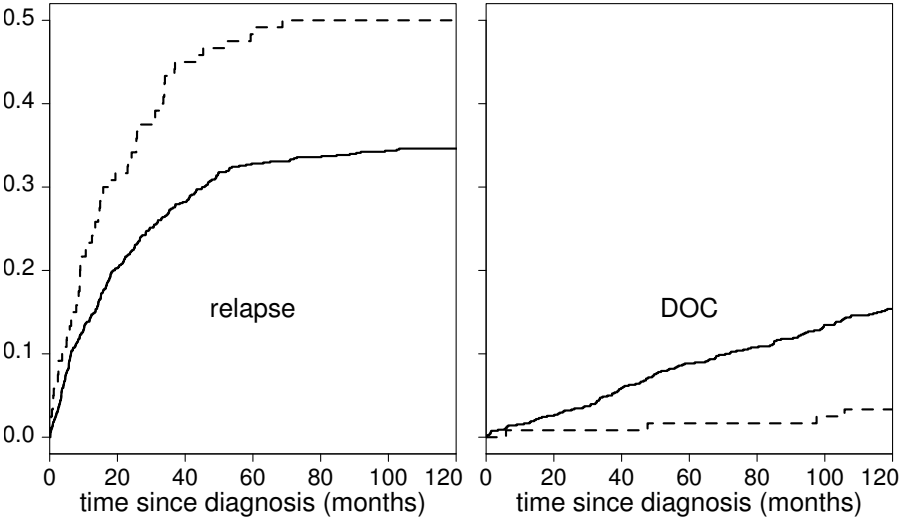
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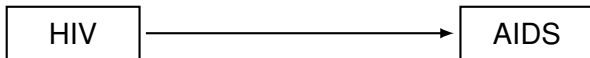
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 - H: Death competing risk of non-fatal event (e.g. disease onset)

H-1: Bladder cancer relapse, death other causes competing

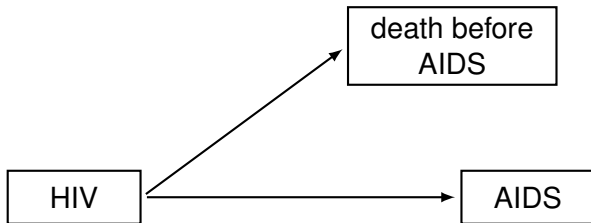


H-2: Death prevents AIDS in HIV infected injecting drug users.



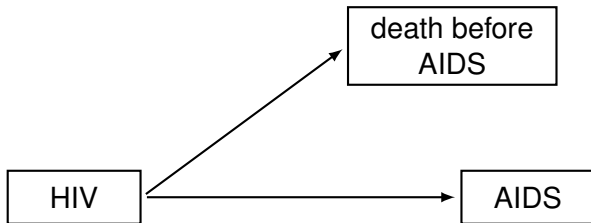
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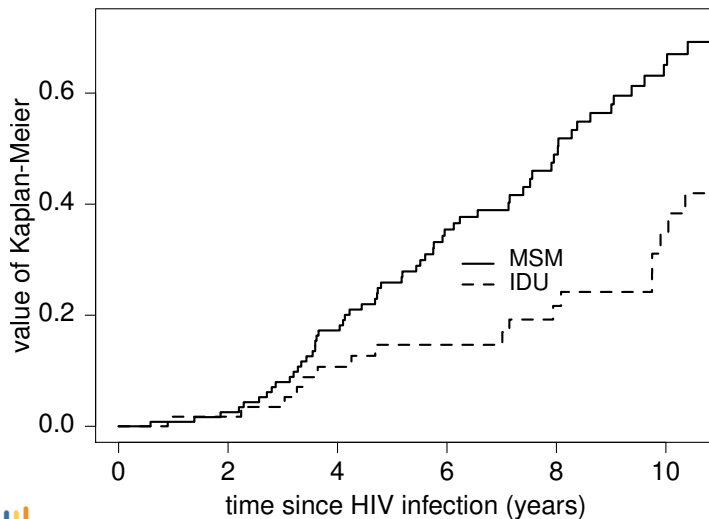
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Data from Amsterdam Cohort Studies: 99 IDU; 127 MSM

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- Compare men who have sex with men (MSM) and injecting drug users (IDU)
Data from Amsterdam Cohort Studies: 99 IDU; 127 MSM
- Kaplan-Meier: leave risk set at death before AIDS
Assumption: those that died had equal AIDS risk

Kaplan-Meier: IDU much slower progression ($p = 0.001$)



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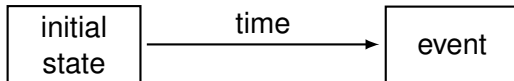
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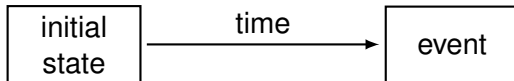
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“We all die, but not all at the same age”



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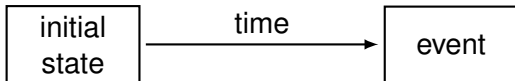
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 1. Multiple event types. Competing risks:
“we all die, but not all at the same age and from the same cause”

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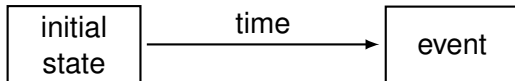
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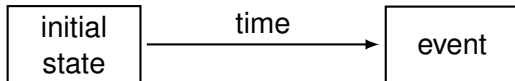
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 - **Events/Transitions.** Initial, intermediate and final states

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 - Events/Transitions. Initial, intermediate and final states
 - **Time**. What is the time origin? Multiple time scales?

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Main quantities in classical survival analysis

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survival function $\bar{F}(t) = 1 - F(t)$:
probability (not) to have experienced the event by time t
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One-to-one relation

Each of the three quantities completely describes distribution:

discrete Event times: $t_1, \dots, t_L$

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(1)

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(2)

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$$\bar{F}(t) = P(T > t) = \prod_{t_l \leq t} \{1 - h(t_l)\}$$

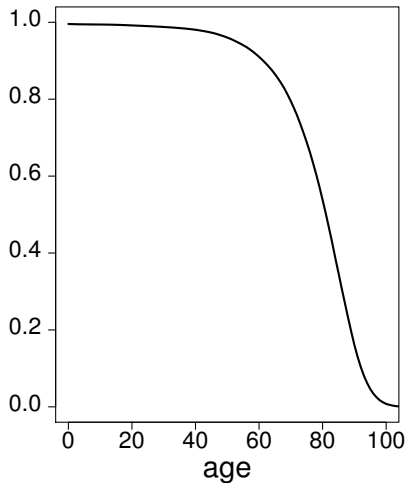
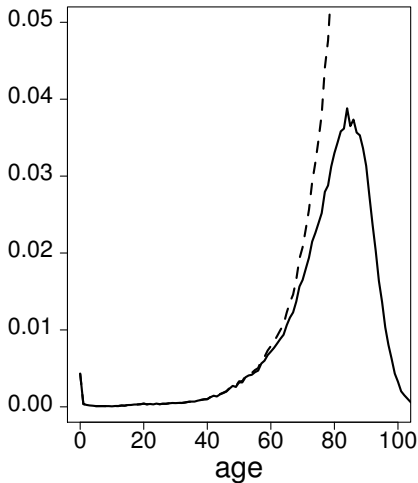
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$$\bar{F}(t) = \exp \left\{ - \int_0^t h(s) ds \right\} = \exp \{ -H(t) \}$$

Mortality by age in Swedish population (Exercise I.5.1)

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Notation and definitions

- Competing risks: type $E \in \mathcal{K}$.

We can assume $\mathcal{K} = \{1, \dots, K\}$

- $T \sim F$ time to event (of any type); $F(t) = P(T \leq t)$
- Overall hazard h :

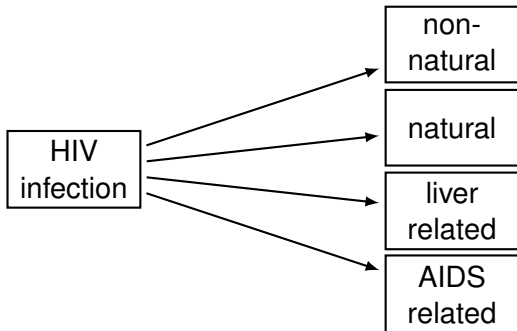
$$\bar{F}(t) = 1 - F(t) = P(T > t) = \exp\{-\int_0^t h(s)ds\}$$

- Cause-specific cumulative incidence:

$$F_k(t) = P(T \leq t, E = k); E \in \mathcal{K}$$

$$\text{Property: } \sum_{e=1}^K F_e(t) = \sum_{e=1}^K P(T \leq t, E = e) = F(t)$$

B: Spectrum in causes of death after HIV infection



- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)
- Side effects of cART? *Etiology* ($\approx$ cause-specific hazard)
- In the end we all die. *Prediction* ($\approx$ subdistribution hazard)

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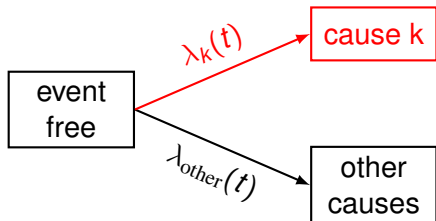
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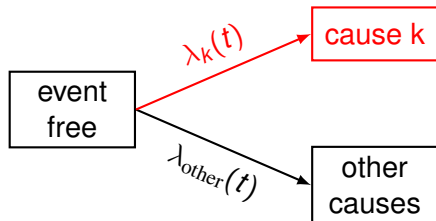
The multi-state approach: cause-specific hazard



- Transition rate to cause k. For continuous distribution:

$$\lambda_k(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t, E = k \mid T \geq t)}{\Delta t}$$

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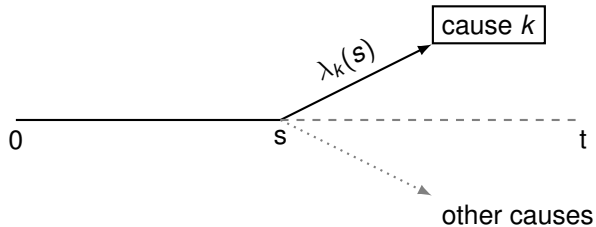


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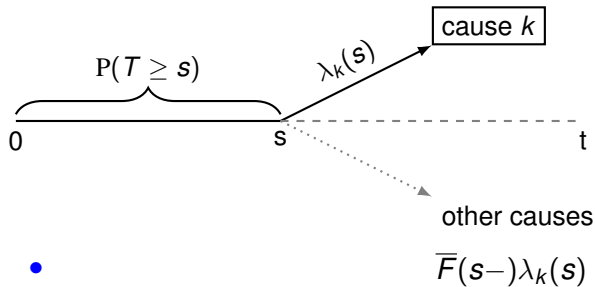
$$\lambda_k(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t, E = k \mid T \geq t)}{\Delta t}$$

- Sum over causes is overall hazard: $\sum_{e=1}^K \lambda_e(t) = h(t)$
- Cause-specific hazard directly generalizes to multi-state setting (called transition hazard)

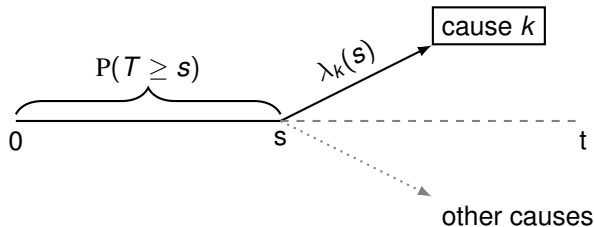
From hazard to cumulative scale $P(T \leq t, E = k)$



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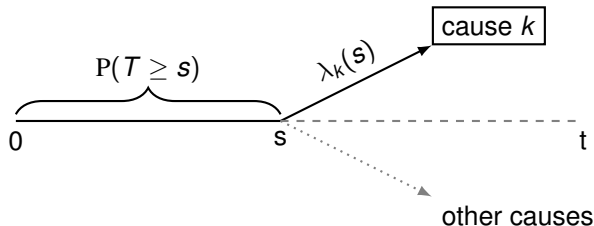


From hazard to cumulative scale $P(T \leq t, E = k)$



- $$F_k(t) = P(T \leq t, E = k) = \int_0^t \bar{F}(s-) \lambda_k(s) ds$$

From hazard to cumulative scale $P(T \leq t, E = k)$



- $F_k(t) = P(T \leq t, E = k) = \int_0^t \bar{F}(s-) \lambda_k(s) ds$
- Depends on all cause-specific hazards via overall "survival"

$$\bar{F}(s) = \exp\left\{-\int_0^s h(u) du\right\} = \exp\left\{-\sum_{e=1}^K \int_0^s \lambda_e(u) du\right\}$$

$\Rightarrow \lambda_k$ does not uniquely specify $P(T \leq t, E = k)$

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Notation and definitions

- Competing risks: type $E \in \mathcal{K}$.
We can assume $\mathcal{K} = \{1, \dots, K\}$
- $T \sim F$ time to event (of any type); $F(t) = P(T \leq t)$
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 $T_k = T \times I\{E = k\} + \infty \times I\{E \neq k\}$ (event of interest)

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Subdistribution hazard

$$\begin{aligned}
 h_k(s) &= \lim_{\Delta s \downarrow 0} \frac{1}{\Delta s} \times P\{s \leq T_k < s + \Delta s \mid T_k \geq s\} \\
 &= \lim_{\Delta s \downarrow 0} \frac{\frac{1}{\Delta s} P\{s \leq T < s + \Delta s, E = k\}}{P\{T \geq s \text{ or } (T < s, E \neq k)\}}
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 - Not a rate in epidemiological sense,

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- Interpretation controversial
 - Not a rate in epidemiological sense, unless we can include immune individuals in the risk set (e.g. vaccination; unobserved cure)
- One-to-one relation with crude risk

$$\overline{F}_k(t) = \prod_{t_l \leq t} \left\{ 1 - h_k(t_l) \right\} \quad \text{or} \quad \overline{F}_k(t) = \exp\left\{ - \int_0^t h_k(s) ds \right\}$$

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Rates and risks in competing risks setting

	hazard		cumulative	
competing risks	marginal	λ^m	net risk	$F^m(t)$
			marginal survival function	
			marginal cumulative incidence	
	cause-specific	λ_k	no corresponding quantity	
	subdistribution	h_k	crude risk	$F_k(t)$
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			overall cumulative incidence	

Exercise I.5.2

Psychiatric patients are treated in two different hospitals, A and B. We want to investigate whether the effect of the treatment on the development of side effects is the same for both hospitals. Hospital A collects information on side effects after discharge, and hospital B does not. In our analysis for hospital A we also use information after discharge, while for hospital B we censor individuals when they are discharged.

Is discharge a competing risk? What type of distribution do we want to quantify? Under which conditions can this be a fair comparison between both hospitals?

Exercise I.5.3

We want to compare the cancer event rates in the virtual situation when the competing risks did not exist. The analysis of the cause-specific hazard models the event of interest in the absence of competing risk events and thus is the appropriate method.

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Reply Latouche *et al.*

*Pintilie's work has the potential to further obscure the issues ... Our main critique concerns the inaccurate assertion: "When modelling the cause specific hazard, one performs the analysis **under the assumption that the competing risks do not exist**".*

Exercise I.5.4

Comment on the following definition of a competing risk:

A competing risk is an event whose occurrence either precludes the occurrence of the event under examination or fundamentally alters the probability of occurrence of that event.

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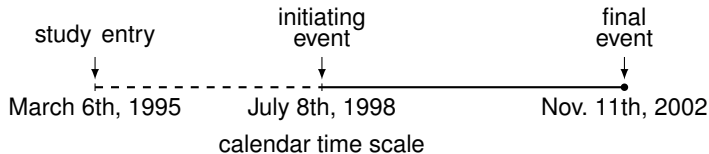
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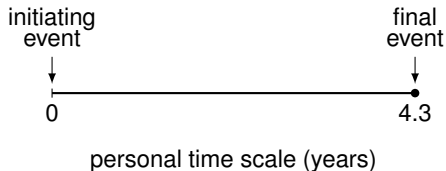
- Setup data analysis

Time scales

- Data collection in calendar time

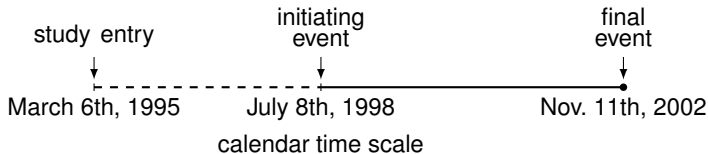


- Often personal/patient time scale more relevant

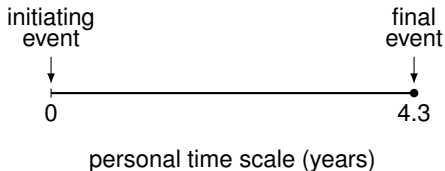


Time scales

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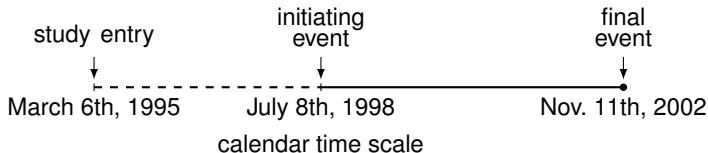
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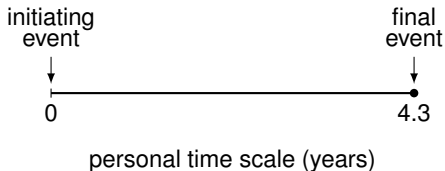
- Every time scale has a reference value: origin $\rightarrow$ end

Time scales

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- Often personal/patient time scale more relevant



- Every time scale has a reference value: origin $\rightarrow$ end
- Classical survival analysis/competing risks: often single principal time scale

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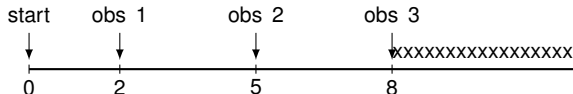
Right censored data

Late entry/left truncation

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The observation process: censored event time

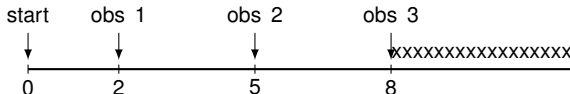
- Right censored: event free until last observation time



event happened after observation time 8

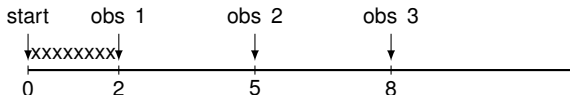
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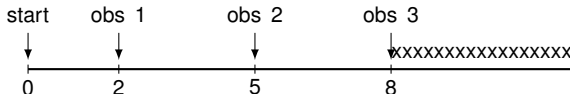
- Left censored: event before first observation time



event happened before observation time 2

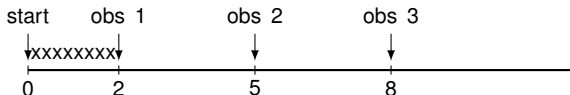
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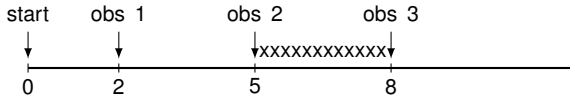
event happened after observation time 8

- Left censored: event before first observation time



event happened before observation time 2

- Interval censored: event between two observation times



event happened between observation times 5 and 8

Right censored event time: three reasons

Administrative censoring At date of analysis. Still in follow-up and event may be observed in the future

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Right censored event time: three reasons

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Competing risks Two types:

Absorbing event: other event (e.g. death) prevents occurrence of event of interest. No further follow-up information

Non-absorbing event: “artificial censoring”. Still in study, but conditions have changed too much to stay included.

Another example: first event analysis

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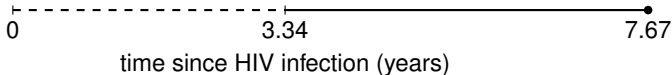
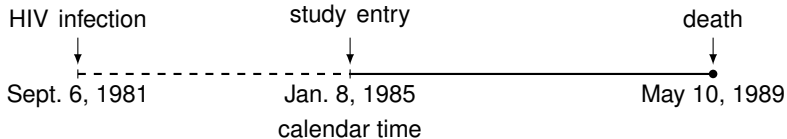
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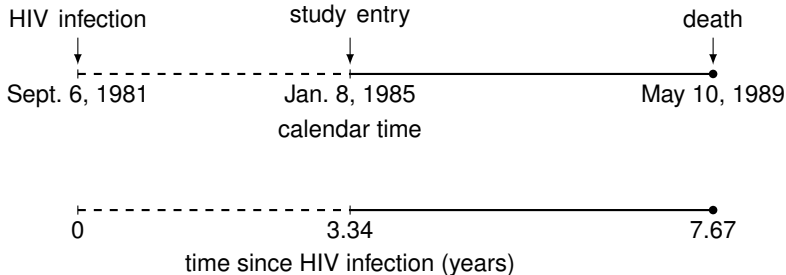
Late entry

Individuals enter risk set after time origin



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Individuals enter risk set after time origin

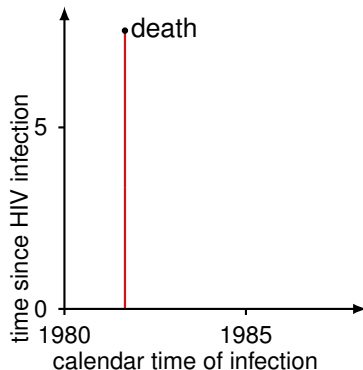


- Late entry usually induces length biased sampling (left truncation)

Left truncation

Fast progressors underrepresented

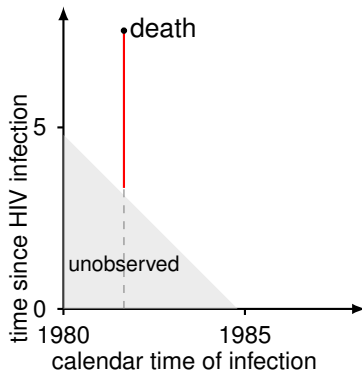
- Example: time from HIV infection to death



Left truncation

Fast progressors underrepresented

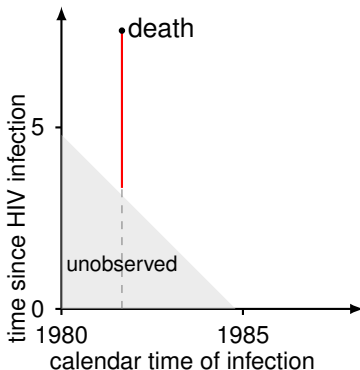
- Example: time from HIV infection to death
- study started in 1985
- individual missed if died before 1985
- individual infected in 1982 only at risk from 1985



Left truncation

Fast progressors underrepresented

- Example: time from HIV infection to death
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- individual infected in 1982 only at risk from 1985
- not same as left censoring



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Probabilistic setup data

- Event time T_i ; may be right censored (administratively, loss to follow-up)

Censoring time C_i

$$X_i = \min\{T_i, C_i\}, \Delta_i = I(T_i \leq C_i)$$

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- Sample: $\{V_i, X_i, E_i\Delta_i\}, i = 1, \dots, N$

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- Left truncation (late entry).
Entry time V_i
- Sample: $\{V_i, X_i, E_i \Delta_i\}, i = 1, \dots, N$
- Assume (V_i, C_i) independent of/non-informative for (T_i, E_i) (possibly conditional on covariables in regression model)

Observed data

$$\{(v_1, x_1, e_1 \delta_1), \dots, (v_N, x_N, e_N \delta_N)\}$$

Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \dots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number “in risk set” (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{iq})$